

Problem 3 Numerical methods for ODEs [10 pts] Consider python code that solves an initial value problem with a Runge-Kutta method and prints the resulting x and y value after N steps:

```

1 def f(x, y):
2     return x * y**2
3
4 def step(x, y, h):
5     k1 = f(x, y)
6     k2 = f(x + h, y + h * k1)
7     y_new = y + (h / 2) * (k1 + k2)
8     x_new = x + h
9     return x_new, y_new
10
11 x = 0
12 y = 0.5
13 h = 0.1
14 N = 20
15
16 for n in range(N):
17     x, y = step(x, y, h)
18
19 print(x, y)

```

- What is the initial value problem that the code solves?
- Write the Butcher tableau for the method used. Is the method explicit or implicit? motivate your answer.
- Compute one step of the algorithm in the code by hand. That is, what is the output if we run the code with $N = 1$ instead?

a) $f(x, y) = xy^2$ $x_0 = 0$
 $y'(x) = xy^2$ $y_0 = 0.5$

$y'(x) = xy^2, y(0) = 0.5$

b) Runge-Kuttz-metode

$$k_i = f(x_n + c_i h, y_n + h \sum_j a_{ij} k_j) \text{ og } y_{n+1} = y_n + h \sum_i b_i k_i$$

Butcher-tableau

c_1	a_{11}	a_{12}
c_2	a_{21}	a_{22}
	b_1	b_2

Les av k_1

Les av k_2

$$k_1 = f(x, y)$$

$$k_1 = f(x_n + 0h, y_n)$$

$$c_1 = 0, a_{11} = 0, a_{12} = 0$$

$$k_2 = f(x + h, y + h k_1)$$

$$k_2 = f(x_n + 1h, y_n + h(1 \cdot k_1 + 0 \cdot k_2))$$

$$c_2 = 1, a_{21} = 1, a_{22} = 0$$

Les av oppdelingen

$$y_{n+1} = y_n + \frac{h}{2} (k_1 + k_2) \Rightarrow b_1 = \frac{1}{2}, b_2 = \frac{1}{2}$$

0	0	0
1	1	0
	1/2	1/2

Eksplicit eller implisitt?

En Runge-Kuttz-metode er eksplisitt hvis hvert nytt k_i bare bruker tidligere k -verdier
 Her bruker k_2 bare k_1 , ikke seg selv $\Rightarrow$ eksplisitt

c) $x_0 = 0, y_0 = 0.5, h = 0.1$

$$f(x, y) = x y^2$$

$$\begin{aligned} k_1 &= f(x_0, y_0) \\ &= f(0, 0.5) \\ &= 0 \cdot (0.5)^2 \\ &= 0 \end{aligned}$$

$$\begin{aligned} k_2 &= f(x_0 + h, y_0 + h k_1) \\ &= f(0 + 0.1, 0.5 + 0.1 \cdot 0) \\ &= f(0.1, 0.5) \\ &= 0.1 \cdot (0.5)^2 \\ &= 0.025 \end{aligned}$$

$$y_1 = y_0 + \frac{h}{2}(k_1 + k_2)$$

$$y_1 = 0.5 + \frac{0.1}{2}(0 + 0.025)$$

$$y_1 = 0.50125$$

$$x_1 = x_0 + h$$

$$x_1 = 0 + 0.1$$

$$x_1 = 0.1$$

Opg 4

x_i	0	2	4
$y_i = f(x_i)$	1	0	1

a) $l_0 = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} = \frac{(x - 2)(x - 4)}{(0 - 2)(0 - 4)} = \frac{x^2 - 6x + 8}{8}$

$$l_1 = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} = \frac{(x - 0)(x - 4)}{(2 - 0)(2 - 4)} = -\frac{x^2 - 4x}{4}$$

$$l_2 = \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} = \frac{(x - 0)(x - 2)}{(4 - 0)(4 - 2)} = \frac{x^2 - 2x}{8}$$

$$p_2(x) = y_0 l_0(x) + y_1 l_1(x) + y_2 l_2(x)$$

$$p_2(x) = \frac{x^2 - 6x + 8}{8} + \frac{x^2 - 2x}{8}$$

$$p_2(x) = \frac{2x^2 - 8x + 8}{8}$$

$$p_2(x) = \frac{1}{4}x^2 - x + 1$$

b) Vi har punktene $(0,1)$, $(2,0)$, $(4,1)$

Vi skal finne Newton-formen til interpolasjonspolynomet

Newton-interpolasjon

$$p_2(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1)$$

$$f[x_i, x_j] = \frac{f(x_j) - f(x_i)}{x_j - x_i}$$

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0}$$

Sett opp punktene

$$x_0 = 0, f(x_0) = 1$$

$$x_1 = 2, f(x_1) = 0$$

$$x_2 = 4, f(x_2) = 1$$

Første delt differanse

$$\begin{aligned} f[x_0, x_1] &= \frac{f(x_1) - f(x_0)}{x_1 - x_0} \\ &= \frac{0 - 1}{2 - 0} = -\frac{1}{2} \end{aligned}$$

Andre førsteordens delt differanse

$$\begin{aligned} f[x_1, x_2] &= \frac{f(x_2) - f(x_1)}{x_2 - x_1} \\ &= \frac{1 - 0}{4 - 2} = \frac{1}{2} \end{aligned}$$

Andreordens delt differanse

$$\begin{aligned} f[x_0, x_1, x_2] &= \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0} \\ &= \frac{\frac{1}{2} - (-\frac{1}{2})}{4 - 0} = \frac{1}{4} \end{aligned}$$

Sett inn i Newton-formen

$$p_2(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1)$$

$$p_2(x) = 1 - \frac{1}{2}(x - 0) + \frac{1}{4}(x - 0)(x - 2)$$

$$p_2(x) = 1 - \frac{1}{2}x + \frac{1}{4}x(x - 2)$$

c) $p_2(x) = \frac{1}{4}x(x - 2) - \frac{1}{2}x + 1 = \frac{1}{4}(x^2 - 2x) - \frac{1}{2}x + 1 = \frac{1}{4}x^2 - x + 1$

d) $p_2(1) = \frac{1}{4} - \frac{1}{2} - \frac{1}{2} + 1 = \frac{1}{4}$

Oppg 8

Vi skal finne vektene w_{-1}, w_0, w_1, w_2

slik at kvadraturregelen $Q[f] = w_{-1}f(-1) + w_0f(0) + w_1f(1) + w_2f(2)$

gir riktig verdi for $\int_0^1 f(s) ds$ for alle polynomer av grad 3

Eksakt for polynomer av grad 3

Da må regelen være eksakt for basispolynomene:

$$1, s, s^2, s^3$$

Bruk $f(s) = 1$

$$\int_0^1 1 ds = [x]_0^1 = 1$$

$$Q[1] = w_{-1} + w_0 + w_1 + w_2 \Rightarrow w_{-1} + w_0 + w_1 + w_2 = 1$$

Bruk $f(s) = s$

$$\int_0^1 s ds = \left[\frac{1}{2} s^2 \right]_0^1 = \left(\frac{1}{2} \cdot 1^2 \right) - \left(\frac{1}{2} \cdot 0^2 \right) = \frac{1}{2}$$

$$Q[s] = w_{-1}(-1) + w_0(0) + w_1(1) + w_2(2) \Rightarrow -w_{-1} + w_1 + 2w_2 = \frac{1}{2}$$

Bruk $f(s) = s^2$

$$\int_0^1 s^2 ds = \left[\frac{1}{3} s^3 \right]_0^1 = \left(\frac{1}{3} \cdot 1^3 \right) - \left(\frac{1}{3} \cdot 0^3 \right) = \frac{1}{3}$$

$$Q[s^2] = w_{-1}(-1)^2 + w_0(0)^2 + w_1(1)^2 + w_2(2)^2 \Rightarrow w_{-1} + w_1 + 4w_2 = \frac{1}{3}$$

Bruk $f(s) = s^3$

$$\int_0^1 s^3 ds = \left[\frac{1}{4} s^4 \right]_0^1 = \left(\frac{1}{4} \cdot 1^4 \right) - \left(\frac{1}{4} \cdot 0^4 \right) = \frac{1}{4}$$

$$Q[s^3] = w_{-1}(-1)^3 + w_0(0)^3 + w_1(1)^3 + w_2(2)^3 \Rightarrow -w_{-1} + w_1 + 8w_2 = \frac{1}{4}$$

Løs

$$\begin{aligned} \begin{aligned} w_{-1} + w_0 + w_1 + w_2 &= 1 \\ -w_{-1} + w_1 + 2w_2 &= \frac{1}{2} \\ w_{-1} + w_1 + 4w_2 &= \frac{1}{3} \\ -w_{-1} + w_1 + 8w_2 &= \frac{1}{4} \end{aligned} \Rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 & | & 1 \\ -1 & 0 & 1 & 2 & | & 1/2 \\ 1 & 0 & 1 & 4 & | & 1/3 \\ -1 & 0 & 1 & 8 & | & 1/4 \end{bmatrix} \xrightarrow{\substack{R_2+R_1 \\ R_3-R_1 \\ R_4+R_1}} \begin{bmatrix} 1 & 1 & 1 & 1 & | & 1 \\ 0 & 1 & 2 & 3 & | & 3/2 \\ 0 & -1 & 0 & 3 & | & -2/3 \\ 0 & 1 & 2 & 9 & | & 5/4 \end{bmatrix} \xrightarrow{\substack{R_3+R_2 \\ R_4-R_2}} \begin{bmatrix} 1 & 1 & 1 & 1 & | & 1 \\ 0 & 1 & 2 & 3 & | & 3/2 \\ 0 & 0 & 2 & 6 & | & 5/6 \\ 0 & 0 & 0 & 6 & | & -1/4 \end{bmatrix} \\ \\ \begin{bmatrix} 1 & 1 & 1 & 1 & | & 1 \\ 0 & 1 & 2 & 3 & | & 3/2 \\ 0 & 0 & 1 & 3 & | & 5/12 \\ 0 & 0 & 0 & 6 & | & -1/4 \end{bmatrix} \xrightarrow{\frac{1}{6}R_4} \begin{bmatrix} 1 & 1 & 1 & 1 & | & 1 \\ 0 & 1 & 2 & 3 & | & 3/2 \\ 0 & 0 & 1 & 3 & | & 5/12 \\ 0 & 0 & 0 & 1 & | & -1/24 \end{bmatrix} \xrightarrow{\substack{R_1-R_4 \\ R_2-3R_4 \\ R_3-3R_4}} \begin{bmatrix} 1 & 1 & 1 & 0 & | & 25/24 \\ 0 & 1 & 2 & 0 & | & 13/8 \\ 0 & 0 & 1 & 0 & | & 13/24 \\ 0 & 0 & 0 & 1 & | & -1/24 \end{bmatrix} \end{aligned}$$

$$\begin{array}{l} R_1 - R_3 \\ R_2 - 2R_3 \end{array} \sim \left[\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 1/2 \\ 0 & 1 & 0 & 0 & 13/24 \\ 0 & 0 & 1 & 0 & 13/24 \\ 0 & 0 & 0 & 1 & -1/24 \end{array} \right] \sim \begin{array}{l} R_1 - R_2 \\ R_2 - R_3 \end{array} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -1/24 \\ 0 & 1 & 0 & 0 & 13/24 \\ 0 & 0 & 1 & 0 & 13/24 \\ 0 & 0 & 0 & 1 & -1/24 \end{array} \right]$$

$$\Rightarrow w_{-1} = w_2 = -\frac{1}{24}, w_0 = w_1 = \frac{13}{24}$$

b) Vi skal vise at Simpsons metode kan skrives som en kombinasjon av trapesmetoden og midtpunktmetoden:

$$S_2[f](a,b) = \frac{1}{3} T_1[f](a,b) + \frac{2}{3} M_1[f](a,b)$$

Trapesmetoden med ett intervall

$$T_1[f](a,b) = \frac{b-a}{2} (f(a) + f(b))$$

Midtpunktmetoden med ett intervall

$$L_2 \ c = \frac{a+b}{2}, \text{ da er } M_1[f](a,b) = (b-a)f(c)$$

Simpsons metode med ett dobbeltintervall

$$S_2[f](a,b) = \frac{b-a}{6} (f(a) + 4f\left(\frac{a+b}{2}\right) + f(b))$$

$$S_2[f](a,b) = \frac{1}{3} T_1[f](a,b) + \frac{2}{3} M_1[f](a,b)$$

$$S_2[f](a,b) = \frac{1}{3} \frac{(b-a)}{2} (f(a) + f(b)) + \frac{2}{3} (b-a)f(c)$$

$$S_2[f](a,b) = \frac{b-a}{6} (f(a) + f(b)) + \frac{b-a}{6} 4f(c)$$

$$S_2[f](a,b) = \frac{b-a}{6} (f(a) + f(b) + 4f(c))$$

$$S_2[f](a,b) = \frac{b-a}{6} (f(a) + f(b) + 4f\left(\frac{a+b}{2}\right))$$

$$S_2[f](a,b) = \frac{b-a}{6} (f(a) + 4f\left(\frac{a+b}{2}\right) + f(b))$$